

# 09 - Backend API Requirements

Visor DOM-to-Agent Context Compiler  
DOC-09 | 0.1 requirements pack | Generated 2026-06-14

This document keeps the MVP local-first while defining a future opt-in backend API surface, security requirements, and suggested architecture.

| Field         | Value                                                                                             |
|---------------|---------------------------------------------------------------------------------------------------|
| Owner         | Rohan PX / Visor project                                                                          |
| Audience      | AI coding agent, reviewer, future contributors                                                    |
| Status        | Draft baseline for MVP build                                                                      |
| Decision rule | MVP is local-first, minimal-permission, Chrome Manifest V3, no backend unless explicitly enabled. |

# Document control

| Item          | Value                                                                                  |
|---------------|----------------------------------------------------------------------------------------|
| Project       | Visor DOM-to-Agent Context Compiler                                                    |
| Document code | DOC-09                                                                                 |
| Generated on  | 2026-06-14                                                                             |
| Primary goal  | Turn raw webpage DOM into safe, structured, agent-ready context.                       |
| MVP boundary  | Chrome extension, local-only compiler, JSON/Markdown export, privacy redaction, tests. |

## 1. Backend stance

MVP requires no backend. A backend is allowed only in later versions for opt-in sync, team workspaces, saved context history, API access, or AI-assisted compilation. Raw page content must never be sent to a backend by default.

## 2. Future backend use cases

- Encrypted user account sync for settings and site profiles.
- Opt-in saved context history.
- Team workspaces and shared extraction profiles.
- Server-side heavy compilation for large pages only after consent.
- API endpoints for tools that want AgentContext programmatically.
- Billing and usage dashboard in a future commercial version.

## 3. API surface draft

| Method | Route                      | Purpose                                     | MVP? |
|--------|----------------------------|---------------------------------------------|------|
| POST   | /api/context/compile       | Compile server-side after explicit consent. | No   |
| POST   | /api/context/save          | Save compiled context to account.           | No   |
| GET    | /api/context/:id           | Fetch saved context.                        | No   |
| DELETE | /api/context/:id           | Delete saved context.                       | No   |
| POST   | /api/site-profiles         | Create or update extraction rule profile.   | No   |
| GET    | /api/site-profiles/:domain | Load domain rules.                          | No   |
| POST   | /api/redact                | Optional server-side redaction service.     | No   |

## 4. Example future request contract

```
POST /api/context/save
Authorization: Bearer <token>
Content-Type: application/json

{
  "schemaVersion": "agent_context.v1",
  "source": { "url": "https://example.com", "title": "Example", "capturedAt": "..."},
  "privacyReport": { "riskLevel": "low", "externalSharingAllowed": true },
  "mainContent": [],
  "tokenProfile": {}
}
```

## 5. Backend security requirements

| ID      | Requirement                                                              | Priority  | Acceptance signal                                       |
|---------|--------------------------------------------------------------------------|-----------|---------------------------------------------------------|
| API-001 | Cloud mode must be opt-in and visibly enabled by the user.               | P0 future | No backend endpoint called unless cloud mode flag true. |
| API-002 | Raw content sync must require explicit save/export action.               | P0 future | Passive compile does not upload content.                |
| API-003 | All backend traffic must use HTTPS.                                      | P0 future | No HTTP endpoint in production config.                  |
| API-004 | User can delete saved contexts permanently.                              | P0 future | Delete endpoint removes context and indexes.            |
| API-005 | Sensitive contexts should support client-side encryption before upload.  | P1 future | Encryption design approved before cloud release.        |
| API-006 | API must separate metadata, content, redaction report, and audit events. | P1 future | DB schema enforces separation.                          |

## 6. Recommended future stack

| Layer          | Recommended option      | Reason                                                                 |
|----------------|-------------------------|------------------------------------------------------------------------|
| API            | FastAPI or Node/Express | Both fit TypeScript/Python ecosystems; choose based on team strength.  |
| Database       | PostgreSQL              | Reliable structured storage for users, profiles, contexts, audit logs. |
| Cache/queue    | Redis                   | Async compilation jobs, rate limits, status.                           |
| Object storage | S3-compatible           | Large context snapshots if cloud save is enabled.                      |
| Auth           | OAuth/email magic link  | Avoid custom password storage early.                                   |